Pre-specified analysis plans

Supplement to: *How fragile a deep lung-nodule detector appears depends on the texture of simulated noise: a paired evaluation of training-time augmentation*

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Code: `https://github.com/Mahmoudyasen/lung-nodule-detector-robustness`

About this document

The analyses of the paper were fixed in three plans before the data they concern were generated or, for the first plan, before the pooled results were seen. This document gives all three:

Part	Plan	Fixed	Evidence
A	Families F1–F3 (blur, realistic noise, paired TCIA data), secondary analyses, descriptive NLST analysis	1 Nov 2026, 11:30 (rev. 1); 20:01 (rev. 2)	git commits `5d5bb18`, `0b2973e`
B	Family F4: noise texture at equal level	2 Nov 2026, 16:58	SHA-256 record
C	Families F5–F7: in-plane blur, noise level, real reconstruction kernels; second training seed	2 Nov 2026, 20:31	SHA-256 record

Times are local (UTC+3). Parts B and C reproduce the plan files word for word; only the layout was changed (Markdown converted to typeset text). The original files are in `plans_original/` of this repository, and their SHA-256 hashes can be checked against the values recorded before any of their data were generated (the record files are in `hash_records/`):

File	SHA-256	Recorded
`texture_test_prespec.md` (Part B)	`7cf2ba56f78e1eb39c2d75cbf0b8d308b4141a13430300e129748ee264162da4`	2 Nov 2026, 16:58:38
`review_prespec.md` (Part C)	`edc7a8be4f3de8df1a1e0f5607220ab0be36fa71499d205c78c918c7384b99fe`	2 Nov 2026, 20:31:58

The record times are file times of the hash files written next to the analysis outputs on the first author's computer; like the git commit times of Part A, they are not external, notarised time stamps. Each plan is followed by notes written afterwards ("Notes"), which list deviations and analyses added after the data were seen; the paper labels these as not pre-specified.

Part A. Plan for families F1–F3

This plan was kept in the first author's project progress document. Section A.2 reproduces its final version (revision 2) word for word; only the layout was changed. Section A.3 lists every difference from the earlier committed version.

A.1 Timeline and evidence

Times are local (UTC+3). "Recorded" means the time is stated in the plan text itself and has no independent record. Git commit times come from the first author's local repository; file times are file-system modification times of the result files. Neither is an external, notarised time stamp.

Time	Event	Evidence
30 Oct, 02:01	Fold-0 results for the image-domain conditions written (seen before the plan was written)	file time
30 Oct	Plan written; revision 1 the same day (projection-domain noise and NLST removed from the scope)	recorded
1 Nov, 11:30:32	Plan (revision 1) first committed	git `5d5bb18`
1 Nov, 11:33	Fold-1 inference starts (first output file for folds 1–3)	file time
1 Nov, ≈18:35	Revision 2 written (projection-domain noise and NLST restored)	recorded
1 Nov, 20:00	Pooled four-fold image-domain results written by the inference script	file time
1 Nov, 20:01:45	Revision 2 committed	git `0b2973e`
2 Nov, 03:20	First TCIA result file	file time
2 Nov, 06:04	First projection-domain-noise result file	file time
2 Nov, 12:28	NLST candidate file	file time
2 Nov, 13:15	Final statistics (`scripts/p2_e1_final.py`)	file time

Full commit hashes:
`5d5bb180fb58d3cdc149e1e8d90843f9b73f3813` (revision 1)
`0b2973ef4cfbdc7881048699bc5730a06ac3a4fb` (revision 2)

What the evidence shows. F1, F3 and the correlated-noise part of F2 were committed before any result for folds 1–3 existed. Revision 2 only added conditions for which no data existed yet: projection-domain noise to F2, and the NLST analysis. Revision 2 was committed about 1.5 minutes (85 s) after the pooled four-fold results were written. Its stated writing time (18:35) is earlier, but there is no independent record of it. Adding three comparisons to F2 made the Holm correction of the four correlated-noise comparisons stricter, not looser.

A.2 The plan (revision 2, as committed)

Fixed before the new data are scored. Only fold 0 (image-domain conditions) has been seen. Everything below is fixed now and will be reported as planned, including null results. Revision (30 Oct, no new result seen): the projection-noise conditions were removed from family F2 and the NLST item was dropped, because tasks 2 and 4 left the paper; nothing else changed. **Revision 2 (1 Nov 2026, ≈ 18:35, while the E1 inference of fold 3 was still running; no pooled 4-fold, projection-noise or NLST result seen):** the full scope is restored, so the projection-domain conditions return to F2 and the NLST check returns as a descriptive secondary analysis (items 2, 3 and 5 below); nothing else changed.

1. **Effect measure.** ΔCPM = CPM(S4) − CPM(S3) on identical perturbed images (paired). LUNA16: pooled over the four test folds (356 scans, each scan scored by its own fold's models). Resampling unit: the scan (LUNA16), the patient (TCIA). Paired bootstrap, 10,000 resamples; 95 % percentile CIs; two-sided $P = 2 \cdot \min(P^*(\Delta \leq 0), P^*(\Delta \geq 0))$.
2. **Primary families**, Holm correction within each family at $\alpha = 0.05$: **F1 kernel**: soft σ 1.0, soft σ 1.5, soft 1.0 + white 60 HU. **F2 realistic noise**: correlated 40/80 HU at 0.7/1.4 mm grain (4) and projection-domain noise at 50 / 25 / 10 % of the TCIA reference (screening) dose (3); 7 comparisons. The projection-noise model (reference I0, tube-current-modulation fit, noise filter per kernel class, slice-thickness scaling) is fixed on TCIA before any LUNA16 projection-noise scan is scored; it is accepted if its held-out per-slice noise-SD error is clearly below that of a constant tube current (pilot, 3 cases: 17 % vs 37 %), otherwise the result is reported and the model revised, still before scoring. **F3 TCIA**: the difference in degradation, $\Delta\Delta = [\mathrm{CPM}_{S4}(10\,\%) - \mathrm{CPM}_{S4}(\mathrm{full})] - [\mathrm{CPM}_{S3}(10\,\%) - \mathrm{CPM}_{S3}(\mathrm{full})]$, and

ΔCPM at 10 %; each averaged over the four fold-model pairs (replicate models on the same 49 patients, not independent samples), bootstrap over patients with all eight models scored on the same resample; primary set as defined above (49 patients, 34 lesions).

3. **Secondary** (reported, unadjusted, labelled exploratory): white noise 20–100 HU (expected null; equivalence margin ±0.02 CPM), sharper kernels, sensitivity at 1/8 FP/scan, per-fold sign consistency of ΔCPM, TCIA solid lesions only, results by vendor and kernel of the original LUNA16 scan.
4. **NLST (descriptive, secondary, no P values)**: S3 and S4 of fold 0 on the fixed stratified sample (1,955 series, one per patient, seed 0, exclusions applied); candidates per scan and score distributions at the frozen threshold, by vendor × kernel class × round; S4 − S3 differences with bootstrap 95 % CIs over patients. NLST has no nodule coordinates, so no sensitivity is claimed.
5. **"White noise is not a dose test"** is shown descriptively: CPM drop from clean under white 80 HU vs correlated 80 HU, with paired CIs; no separate test.
6. **Fixed choices that are not tuned on test data:** score threshold 0.02 and max 100 candidates/scan (as frozen); the TCIA primary set and the handling of the unresolved lesions (above).

TCIA primary set ("defined above" in item 2; same section of the progress document, same commit). Fixed rule (simplified 30 Oct, instead of recovering the slices by hand): a lesion with an unknown slice is left out of the reference, and any detection within its radius in-plane, at any slice, is ignored (the LUNA16 "excluded finding" rule applied as a column), so it counts neither as a hit nor as a false positive; patient C128 is excluded entirely because its position error could affect both lesions. Primary set: 49 patients, 34 lesions (22 solid, 12 subsolid; diameter 5.4–25.1 mm, median 8.0 mm).

A.3 Changes between the committed versions

Revision 1 (`5d5bb18`) differs from revision 2 (`0b2973e`) only in: (i) the heading and the sentence on revision 2; (ii) F2 was "**F2 correlated noise**: correlated 40/80 HU at 0.7/1.4 mm grain (4)." (no projection-domain conditions); (iii) item 4 (NLST) was absent. The plan text was not changed after revision 2 (identical in commit `20403ab`, 2 Nov).

A.4 Terms used in the paper

"Kernel" in this plan means the Gaussian blur and unsharp masking applied to the images; the paper calls them blur and sharpening, and keeps "kernel" for real reconstruction kernels. "Soft σ 1.0 / 1.5" are in voxels (0.70 / 1.05 mm in-plane). "E1" is the image-domain perturbation test. S3 and S4 are the models without and with augmentation. Analyses added after this plan are labelled as not pre-specified in the paper.

Part B. Plan for family F4 (noise texture)

File `texture_test_prespec.md`, reproduced word for word.

P2 task 5: noise-texture test (pre-specified)

Written 2 Nov 2026, ~17:15, before any NPS-matched noise was generated or scored. The SHA-256 hash of this file is recorded in `analysis/p2_texture/PRESPEC_SHA256.txt` before the generation step. Any later change to this plan is listed at the end, with the date and the reason.

Why

In the draft, the claim that noise texture explains the different effects of white, correlated and projection-domain noise is argued but not tested. The three noise models differ in texture, level, spatial uniformity and insertion path all at once. For example, "80 HU" image-domain noise adds 67 HU over the lung crop, while projection noise at 10 % dose adds a median of about 102 HU. This test holds the level and the insertion path fixed and varies only the texture.

Conditions (LUNA16 test folds 0-3, 356 scans; S3 and S4 of each fold; paired, identical inputs)

All noise is added to the ORIGINAL full-field LUNA16 image (HU, native grid). It then goes through the unchanged preprocessing with the same lung mask, exactly the path used for the projection-domain noise (`scripts/17_luna_projnoise.py`). Seeds are fixed per (scan, condition). The noise is stationary, zero-mean, Gaussian and additive.

- **T1 `tex_nps`**: in-plane noise power spectrum (NPS) equal to the measured NPS of the real TCIA noise (10 %-dose minus routine image). The B50f chest pairs give the spectrum for sharp-kernel scans and the B30f abdomen pairs for soft-kernel scans, the same kernel-class split as the projection noise. Slices are independent along z, as in the projection noise. The radial NPS is measured in cycles/mm and applied at each scan's own pixel spacing. It is zero above the highest frequency measured on TCIA.
- **T2 `tex_gauss`**: the texture of the correlated image-domain noise of the draft: white noise low-pass filtered with a 3D Gaussian of $\sigma = 1.41$ mm in-plane and 2.5 mm along z (2 voxels on the detector grid; the "1.4 mm grain" condition), here in physical units on the native grid.
- **T3 `tex_white`**: white noise, independent per native voxel.

Level: for every scan and condition, the added-noise SD over the preprocessed lung crop (preprocessed noisy volume minus the clean archive) is matched to that scan's projection-domain noise SD at 10 % dose (`noise_sd_added_hu` in `projnoise/a0.1/.../<uid>.json`). The scale is set in two passes; the accepted tolerance is ±3 % per scan, and any scan outside it is reported. T1-T3 and proj0.1 therefore have the same level, crop and insertion path. They differ in texture (T1 vs T2 vs T3) or, for T1 vs proj0.1, only in the spatial non-uniformity and streak structure of the projection noise.

Known-answer checks before scoring

1. The generated T1 noise, measured with the same ROI method, reproduces the target radial NPS shape (normalised L1 distance < 0.05). The generated T2 and T3 noise have the expected flat or Gaussian shape.
2. With zero amplitude, the output is byte-identical to the clean archive (3 scans).
3. Every scan's achieved SD is within ±3 % of its target.
4. The new inference path reproduces the existing proj0.1 candidates of one scan exactly.

Primary family F4 (Holm within the family, $\alpha = 0.05$, m = 2)

For each model M ∈ {S3, S4}: contrast CPM_M(T1) − CPM_M(T2), on the same paired bootstrap as F1-F3 (10,000 resamples over scans, common resamples, two-sided P, floor 2/10,000). The texture hypothesis predicts > 0: at equal level and path, noise with the real texture harms less than Gaussian-blob noise.

Pre-specified equivalence check (secondary, not in Holm)

For each model, CPM_M(T1) − CPM_M(proj0.1), 95 % CI inside ±0.02 → "stationary noise with the real NPS reproduces the projection-domain result".

Secondary (reported, unadjusted)

- Each model's drop from clean under T1, T2 and T3 (paired CIs).
- CPM_M(T3) − CPM_M(T1).
- ΔCPM S4 − S3 per condition.
- The above by native kernel class.
- Candidates per scan.
- Measured TCIA noise correlation between adjacent slices (descriptive; checks the z-independence assumption).

Interpretation rule (fixed now; the paper's wording follows it)

- **Supported**: F4 significant for both models AND the equivalence check holds for both. Texture at equal level explains most of the difference, and stationary NPS-matched noise suffices to simulate dose.
- **Partly supported**: F4 significant for at least one model, but the equivalence check fails. Texture matters, but so does the non-stationary or streak structure of projection noise; NPS-matched stationary noise is not a full substitute.
- **Not supported**: F4 significant for neither model. At equal level and path, texture does not explain the difference; the draft's explanation must be withdrawn.

The result is reported whichever way it goes.

Changes after this point

(none yet)

Notes on Part B (written afterwards)

File `texture_test_prespec_NOTES.md`, reproduced word for word.

Notes on the noise-texture pre-specification

The pre-specification (`texture_test_prespec.md`) is left unchanged so that its SHA-256 hash stays valid. Notes added afterwards:

1. **Time stamp.** The file says "Written 2 Nov 2026, ~17:15". Its hash was recorded at 16:58:38 on 2 Nov 2026 (the hash file written next to the analysis outputs), so the file was written before 16:58. "~17:15" was an estimate written into the text and is wrong. The order of events (plan → hash → generation 17:xx → scoring) is unaffected.
2. **Level tolerance.** 16 of the 356 scans × 3 textures were outside the ±3 % tolerance after two passes (real NPS 5, white 11, Gaussian grain 0; largest deviation 5.0 %). As the plan requires, they are reported, and they were kept in the analysis.
3. **Result.** Verdict by the pre-specified rule: "partly supported" (F4 significant for both models; equivalence with projection noise failed for both). See `texture_results.json` written by `scripts/28_texture_stats.py`.

Part C. Plan for families F5–F7 and the second seed

File `review_prespec.md`, reproduced word for word.

P2 review experiments (pre-specified): blur geometry, noise level, real kernels, second seed

Written 2 Nov 2026, before any of the data below was generated or scored. The SHA-256 hash of this file is recorded in `analysis/p2_review/PRESPEC_SHA256.txt` before the first generation step; the time of that record is the time of this plan. Any later change is listed at the end, with the date and the reason. These analyses answer six review points on draft v2 (2 Nov). They are additional to the plan of 30 Oct and to the texture plan (`p2_texture_prespec.md`); none of the earlier results is changed, and each family below has its own Holm correction. Only eligibility counts were looked at before writing (NLST: 29,376 eligible exams), no outcome.

Common to all: same detectors (S3, S4 of folds 0-3; frozen files read only), same inference code and threshold (0.02), same scorer (LUNA16 protocol, CPM), paired bootstrap with 10,000 common resamples over scans (LUNA16) or patients (TCIA, NLST), two-sided P with floor 2/10,000, percentile 95 % CIs, $\alpha = 0.05$.

R1. Blur geometry (review point 2): in-plane blur (family F5)

The E1 blur is isotropic in voxels on the 1.25 × 0.703 × 0.703 mm grid: $\sigma = 1.0$ / 1.5 voxels is 1.25 / 1.88 mm along z and 0.70 / 1.05 mm in-plane. Reconstruction kernels act in-plane. New conditions, applied exactly like the E1 blur (preprocessed volume, same per-scan loading, separable Gaussian with reflect padding) but along y and x only:

- `xy1.0`: $\sigma = 1.0$ voxel in-plane (0.70 mm), none along z.
- `xy1.5`: $\sigma = 1.5$ voxels in-plane (1.05 mm), none along z.
- `xycal` (secondary): in-plane σ equal to the median calibrated σ of R3 (in mm, converted to voxels), rounded to 0.05 mm. If it equals 0.70 or 1.05 mm within 0.05 mm, it is still run and reported.

F5 (Holm, m = 2): ΔCPM = CPM(S4) − CPM(S3) at `xy1.0` and at `xy1.5`, LUNA16 folds 0-3 pooled (356 scans). Secondary: each model's loss from clean; ΔCPM at `xycal`; the loss under in-plane blur relative to the loss under the 3D blur of the same σ. The text reports σ in mm everywhere from now on.

R2. Noise level (review point 3): level response of the three textures (family F6)

The texture test matched the level of all textures to each scan's projection-noise SD at 10 % dose (factor 1). New levels: factors 0.5 and 2 of the same per-scan target, for `tex_nps`, `tex_gauss` and `tex_white` (6 conditions), made with the unchanged generator (same noise fields and seeds as factor 1, only the amplitude differs; two-pass matching, ±3 % tolerance, deviations reported). Same insertion path and inference as the texture test.

F6 (Holm, m = 2): for each model, CPM(`tex_nps`, factor 2) − CPM(`tex_gauss`, factor 0.5). Prediction > 0: real-NPS noise at four times the SD (16 times the variance) still harms less than Gaussian-grain noise at the lower level. Wording rule: significant for both models → "the spectrum mattered more than a four-fold change in level"; for one or neither → the paper keeps "at equal level" and states that a four-fold level change was (partly) able to offset the spectrum. Secondary: CPM against level (0, 0.5, 1, 2) per texture and model; the same contrast for white noise at factor 2 minus Gaussian grain at factor 0.5; ΔCPM S4 − S3 at each condition.

R3. Real reconstruction kernels in NLST (review points 4 and 6) (family F7)

Sample. NLST exams with two axial reconstructions of the same acquisition: the usable archive series (primary; exclusions applied) with kernel class soft or sharp, and a partner series of the same study with the other class, the same slice thickness and the same number of images. Thickness ≤ 2.5 mm (the LUNA16 range). If several partners qualify, the one with the lowest series number. One exam per patient; shuffle with seed 0; up to 125 exams per vendor (GE, Philips, Siemens, Toshiba): 500 exams, 1,000 series. **Conversion.** Both series from raw DICOM to detector spacing, the LUNA16 path (`convert_native`), each cropped with the primary series' archive lung box, so the two volumes cover the same world region. A pair is kept only if the

two volumes have the same shape, origin and spacing; the number dropped is reported. Conversion failures are reported. **Inference.** All eight fold models (S3, S4 of folds 0-3) on both reconstructions. **Outcome (no nodule positions in NLST).** For each model and exam, confident detections (score ≥ 0.5) on the soft and on the sharp reconstruction are matched one-to-one (greedy by score, centre distance ≤ 5 mm). Kernel agreement of a model = $2 \times$ matched / (confident soft + confident sharp), pooled over exams (ratio of sums); exams without confident detections on either side contribute nothing. **F7 (m = 1):** $\Delta A = A(S4) - A(S3)$, each averaged over the four fold models, bootstrap over patients. Prediction > 0. Wording rule: significant $\rightarrow$ "augmentation made detections more consistent across real kernels"; not significant $\rightarrow$ "augmentation did not measurably change the consistency across real kernels", with the CI. **Secondary (descriptive, CIs, no P):** A per model and vendor; persistence = share of confident detections on one kernel that have any candidate (≥ 0.02) within 5 mm on the other, in both directions; mean score difference (soft − sharp) of matched candidates; candidates per scan by kernel; the same with fold-0 models only. **Kernel calibration (descriptive).** Per exam, the in-plane Gaussian σ (mm; grid 0-3 mm, step 0.05) that, applied to the sharp volume, minimises the mean squared difference to the soft volume over the cropped volume (windowed, normalised as for the detector). Reported: median and IQR per vendor and overall. The overall median defines `xycal` (R1). **Paper.** This analysis replaces the current NLST section as the screening-data result; the descriptive single-series analysis moves to the supplement with a short paragraph.

R4. Second training seed (review point 5)

Two new fold-0 models, trained with the unchanged code and configs and seed 1: S3 (`configs/stage1_retinanet.yaml`, `fold0/seed1_s3_noaug`) and S4 (`configs/stage1_retinanet_aug.yaml`, `fold0/seed1_s4_aug`). The frozen `fold0/seed0` is never written. Both are trained after the sampler fix, so the seed-1 S3 also answers the sampler-fix caveat. Evaluated on the fold-0 test scans (89) under all E1 conditions, projection noise (3 doses), the three textures at factor 1, `xy1.0` / `xy1.5`, and on TCIA (both doses). **Reported (secondary, no new primary family):** (i) seed-to-seed difference in CPM of each configuration per condition (seed 1 − seed 0) with paired CIs; (ii) ΔCPM S4 − S3 of the seed-1 pair next to the seed-0 pair; (iii) replication check: the seed-1 pair's ΔCPM at blur σ 1.0 and 1.5 is positive with a CI excluding zero (stated as replicated or not); (iv) sensitivity analysis of F1-F3: fold 0's S3 replaced by the seed-1 S3 (post-fix), everything else unchanged.

Interpretation and paper (review points 1 and 6)

Title and abstract are rewritten after R1-R4, following the wording rules above. Claims are limited to what was tested: Gaussian blur is called a model of softer kernels; "spectrum more than level" is used only if F6 supports it; the real- kernel claim follows F7 and the calibration. Display items stay at 8 (NLST real kernels replace Fig. 5).

Changes after this point

(none yet)

Notes on Part C (written afterwards)

File `review_prespec_NOTES.md`, reproduced word for word.

Notes on the review pre-specification (`review_prespec.md`)

The plan file is unchanged (SHA-256 in `analysis/p2_review/PRESPEC_SHA256.txt`). Everything below was decided after the data of the step concerned had been generated or scored, and is labelled "not pre-specified" in the paper.

Execution (as planned)

- R3 sample: 500 exams (125 per vendor). Conversion: 0 failures. Geometry check: 102 pairs dropped (Philips 80, Siemens 14, GE 7, Toshiba 1), almost all with a different shape/origin (different reconstruction field of view); 398 exams analysed (GE 118, Philips 45, Siemens 111, Toshiba 124). Calibration: median 0.50 mm (IQR 0.45-0.55), no exam at the grid edge; `xycal` = 0.50 mm = 0.711 voxels in-plane.

- R1, R2, R3 primary results as pre-specified: F5 both significant, F6 both significant (wording rule: "the spectrum mattered more than a four-fold change in level"), F7 significant.

Additions after seeing the data (3 Nov)

1. **R2 level shortfall at factor 2.** The two-pass level matching measures the SD over the preprocessed lung crop, after the clip to [-1000, 400] HU. At factor 2 (median target 204 HU) the clip limited the achievable level: 138 (real NPS), 156 (white) and 88 (Gaussian grain) of 356 scans fell outside +-3 % (largest -17.7 %); the achieved median was 1.9 % (ratio of medians, real NPS) below the target. At factors 0.5 and 1 all or nearly all scans were within tolerance. Sensitivity analysis (not pre-specified; `32_review_stats.py levels`, `levels_sens_within3pct.csv`): F6 restricted to the 218 scans whose real-NPS x2 noise was within +-3 % (all Gaussian x0.5 scans were). Result: the contrast REVERSED, S3 -0.054 [-0.094, -0.012], S4 -0.061 [-0.099, -0.020]. See point 3 for the interpretation. Exploration (scratch, then made reproducible as `32_review_stats.py levelskernel` -> `levels_by_kernel.csv`): the 138 scans outside tolerance are 136 sharp-kernel scans (median x2 target 341 HU vs 164 HU for the kept scans), so the restriction is in effect a restriction to soft-kernel scans. F6 by native kernel class: sharp (151 scans, B50f spectrum) S3 +0.177, S4 +0.165; soft (205 scans, B30f spectrum) S3 -0.045, S4 -0.061. Splitting at the median target level gives the same pattern. A level shortfall of about 2 % cannot explain a swing of this size. Caveat: kernel class is confounded with absolute level (median x2 target 327 HU sharp vs 160 HU soft, because the target follows each scan's projection noise), so the split cannot attribute the difference to the spectrum; the paper says so.
2. **R3 by-vendor CIs.** The plan lists "A per model and vendor" as secondary without fixing the estimator. Added `32_review_stats.py nlstvendor`: the F7 estimator (mean over fold models of S4 - S3) within each vendor, bootstrap over that vendor's exams (10,000, seed 0); known-answer check: the "all" row reproduces F7 exactly (estimate and CI).
3. **Interpretation of F6.** The pre-specified result (positive for both models) is reported as such, with its rule-based wording, but the paper states next to it that the average hides opposite results by kernel class: the sharp-kernel spectrum (at a higher absolute level) beat a four-fold level change, the soft-kernel spectrum did not; spectrum and absolute level cannot be separated in this split. The abstract, introduction, discussion and conclusion no longer claim that texture matters "more than level" in general. F4 (equal level) holds in both kernel classes (soft +0.44/+0.40, sharp +0.68/+0.68), so the texture headline (title) stands.

Paper

- Figure 2c shows blur against in-plane sigma in mm (3D and in-plane-only); Figure 2e shows the level response (replaces the factor-1 texture panel, whose numbers stay in Table 3 and Supplementary Table S6). Figure 5 is the NLST real-kernel analysis; the single-series NLST analysis moved to Supplementary Note 1 and Table S2.

Second review round (3 Nov, after an outside reading of draft v3)

4. **R3 threshold sensitivity (not pre-specified).** S4 scores are shifted slightly lower, so agreement at the fixed 0.5 threshold could change through calibration alone. `32_review_stats.py nlstthr` -> `nlst_pairs_threshold.csv`: same estimator and bootstrap as F7 at thresholds 0.3/0.5/0.7, plus a count-matched variant (each S4 fold model thresholded so that its total confident detections over all series equal those of S3 of the same fold at 0.5; S4 thresholds 0.50-0.62, fold 1 0.22). Known answer: the 0.5 row reproduces F7. Result: +0.023 / +0.021 / +0.023 and +0.020 [0.011, 0.029] count-matched; all fold differences positive in every variant. Confident detections per series at 0.5: S3 2.85, S4 2.76.
5. **Shortfall by kernel class.** `levelskernel` now also writes the median and 90th-percentile shortfall of the real-NPS x2 level (sharp 4.4 % / 7.2 %, soft 1.0 % / 1.7 %); estimates and CIs unchanged (checked against the previous file). The shortfall biases the sharp-kernel F6 contrast towards positive but, on the level curve, is worth far less than its size; the abstract no longer states the kernel-class direction.
6. Text corrections: the sentence "augmentation did not change the loss under real-NPS noise at any level" was false (x2: +0.036 [0.009, 0.059]) and is replaced; the Discussion's "87 % reproduced" (a Dice-style agreement) is replaced by the persistence; "consistently across manufacturers" removed (single fold pairs negative for GE, Philips, Toshiba).

R4 second seed (5 Nov, executed as planned) Queue finished 05:48 (S3 seed 1 trained 3 Nov 11:44 - 4 Nov 06:30, S4 seed 1 4 Nov 06:30 - 5 Nov 01:10, evaluation 01:10 - 05:48). `32_review_stats.py seed`

-> seed_contrasts.csv, seed_sensitivity_f1f3.csv, seed_results.json. (iii) Replication (pre-specified criterion, both strengths): NOT met in full. sigma 1.5 voxel (1.05 mm): seed 1 +0.077 [0.027, 0.126] (seed 0 +0.103); sigma 1.0 (0.70 mm): +0.027 [-0.008, 0.064] (seed 0 +0.048). Paper: "met only in part". (i) Seed differences: clean 0.019 (S3) / 0.004 (S4); median over conditions 0.017 / 0.010; max 0.089 / 0.037, the S3 maximum under correlated noise 80 HU / 1.4 mm (seed-1 S3 more robust, +0.089 [0.045, 0.137]). (ii) Seed-1 pair: S4 - S3 negative under correlated 80 HU noise (-0.062, -0.075) and Gaussian grain (-0.048). (iv) Sensitivity (fold-0 S3 replaced by seed-1 S3): F1 all still significant (P_Holm 0.0006); F3 unchanged (ns); F2 correlated 80 HU / 1.4 mm no longer significant (+0.022 [-0.003, 0.044], P_Holm 0.64; main +0.042, P_Holm 0.0014). The two S3 models differ in seed AND sampler fix; the analysis cannot separate them. Paper: Results (correlated-noise sentence, seed section), Discussion (noise benefits not robust; blur benefit reproduced at 1.05 mm, CI includes zero at 0.70 mm) and limitations rewritten from these numbers. Technical: make_numbers.py sensitivity keys for the two F3 rows renamed (rv.sens.F3DD, rv.sens.F3Dlow) because the raw condition names contain characters that break LaTeX csnames. Paper after the seed results (5 Nov, outside reading): abstract now states that the correlated-noise benefit did not survive retraining a baseline and limits the blur claim to strong blur; Methods state the replication criterion and the sensitivity analysis; Table 3, Fig. 3 and Table S1 flag the correlated-noise 80 HU / 1.4 mm result with a dagger; the seed-1 in-plane 0.70 mm result (-0.008, no benefit) is stated; the 0.089 seed difference is compared with the seed-0 benefit on the same 89 scans (+0.033), not with the pooled effect; real-NPS / white x2 benefits marked unreplicated.